

Hominis Placenta modulates PTSD-like behaviors in SPSS-induced PTSD mice: Regulating energy metabolism and neuronal activity

Mudan Cai, Hee Ra Park, Eun Jin Yang^{*,1}

Department of KM Science Research Division, Korea Institute of Oriental Medicine (KIOM), 1672 Yuseong-daero, Yuseong-gu, Daejeon 34054, South Korea

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ABSTRACT

The symptoms of post-traumatic stress disorder (PTSD) include re-experiencing trauma, avoidance behaviors, negative alterations in cognition and mood. However, the underlying molecular mechanisms are unclear. Dysfunction of hypothalamic-pituitary-adrenal axis (HPA-axis) and dysregulation of glutamatergic and GABAergic systems were shown during PTSD. Therefore, regulating hormonal change or glutamate energy metabolism are considered as a therapeutic approach to alleviate this condition. Herbal medicine may be effective in treating PTSD due to its ability to target multiple underlying mechanisms with various compounds. Hominis placenta (HP) is a traditional medicine widely used in East Asia for various conditions. However, the effect on PTSD has not been clarified. We aimed to investigate the effects of HP treatment in single-prolonged stress with shock (SPSS)-induced PTSD mice and explore its possible mechanisms. HP treatment at ST36 acupoints, combined with herbal medicine and acupuncture point stimulation, was applied three times/week for 2 weeks. HP treatment effectively alleviated anxiety and cognitive decline in SPSS-induced PTSD mice, as detected by Open field and the Y-maze test. Additionally, HP decreased the corticosterone levels and proinflammatory cytokines in the serum, modulated brain energy metabolism, and inhibited glutamate excitotoxicity, while regulating neuronal activity through modulating brain-derived neurotrophic factor (BDNF) levels, as demonstrated by western blot and immunohistochemistry, and flow cytometry analyses. These findings reveal that HP treatment effectively alleviates PTSD-like behaviors by regulating energy metabolism and neuronal activity through modulation of the HPA-axis and BDNF levels in PTSD mice, indicating that HP treatment is a promising therapeutic approach for PTSD.

1. Introduction

Post-traumatic stress disorder (PTSD) is a mental disorder that develops following traumatic events, such as rape, war, violence, or natural disaster. Symptoms of PTSD include intrusive re-experiencing of the past trauma, avoidance behaviors, negative alterations in cognition and mood following a traumatic event in vulnerable populations [1]. At present, selective serotonin reuptake inhibitors (SSRIs) are the primary treatment for patients with PTSD [2]. However, the discontinuation of SSRI treatment in patients with PTSD has been linked to limited efficacy and symptom exacerbation compared to extended treatment, as well as

SSRIs are higher rates of adverse events like insomnia, diarrhea, and nausea. PTSD is associated with several pathological mechanisms, including dysfunctions of the hypothalamic-pituitary-adrenal (HPA)-axis, inflammation, altered metabolism, and perturbations in neurotransmitter systems [3]. Additionally, glucocorticoids (GCs) produced by the HPA-axis strongly influence immune cells, including monocytes, macrophages, and dendritic cells [4]. Compared to healthy controls, patients with PTSD show significantly increased levels of inflammatory markers [5]. Additionally, patients with PTSD often have elevated body mass index and glucose, insulin, and creatinine levels [6], suggesting PTSD-related metabolic dysregulation. Moreover,

Abbreviations: ACTH, adrenocorticotrophic hormone; ATP, adenosine triphosphate; BDNF, brain-derived neurotrophic factor; CAM, complementary and alternative medicine; CORT, corticosterone; GCs, glucocorticoids; GR, glucocorticoid receptor; HP, Hominis placenta; HPA, hypothalamic-pituitary-adrenal; IL, interleukin; OFT, Open field test; OXPHOS, oxidative phosphorylation; PTSD, post-traumatic stress disorder; RT-qPCR, real time-quantitative polymerase chain reaction; SPSS, single-prolonged stress with shock; SSRIs, selective serotonin reuptake inhibitors; TNF, tumor necrosis factor.

* Corresponding author.

E-mail address: yej4823@gmail.com (E.J. Yang).

¹ ORCID: 0000-0002-2963-3923

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dysregulation of the glutamate metabolism leads to disturbances in the balance of glutamate and GABA neurotransmitter associated various disorders, such as mental disorders and neurodegenerative disease. Furthermore, excessive glutamate in the extracellular space induces glutamate excitotoxicity leading to neuronal damage and weakness. Therefore, regulating energy metabolism to maintain proper levels of glutamate and GABA is considered as a therapeutic target for alleviating these diseases. Furthermore, dysregulation of neurotransmission has been implicated in the PTSD, particularly through the activation of NMDA-type glutamate receptor (NMDAR) on the synaptic plasticity.

Hominis placenta (HP) is dried human placenta, widely used as a traditional medicine in East Asia. HP contains peptides, amino acids, enzymes, and immune substances, such as interferons, blood coagulation factors, and hormones and their precursors [7,8]. HP has been widely used for alleviating osteoporosis, promoting the regeneration of injured nerves, relieving rheumatoid arthritis symptoms, and addressing dementia symptoms, especially memory loss, agitation, and disorientation [7,9]. In addition, HP decreases the proinflammatory cytokine levels and oxidative stress and increases brain-derived neurotrophic factor expression and dendrite density around $\alpha\beta$ plaques in Alzheimer's disease (AD) animal models [8,10]. Moreover, HP treatment has shown effectiveness in stress-induced alopecia areata, as presented in a case report [11]. Despite previous studies having demonstrated the anti-inflammatory properties and cognitive protection offered by HP, its effects on PTSD have not yet been investigated. Thus, in this study, we aimed to investigate the effects of HP treatment in single-prolonged stress with shock (SPSS)-induced PTSD mice and explore its possible mechanisms of action related to the HPA-axis.

2. Material and methods

2.1. Animals

Adult male C57BL/6 N mice (21–23 g, 8 weeks old) obtained from Dae Han Biolink (Eumseong-gun, Chungcheonbuk-do, South Korea) were used. The mice were housed under specific pathogen-free control conditions and maintained at standard conditions of temperature ($21 \pm 3^\circ\text{C}$) and humidity ($50 \pm 10\%$) with a 12-h light/dark cycle and provided *ad libitum* access to water and food. The experimental protocol was approved by the Korea Institute of Oriental Medicine (protocol number: 23–011).

2.2. SPSS-induced animal model

PTSD was induced by SPSS as previously described [12]. Briefly, mice were subjected to a 4-h restraint, followed by forced swimming for 20 min, with a subsequent 15-min rest period, following which they were anesthetized using isoflurane [13]. After 1 h, foot shocks (1 mA, 4 s, two times) were directly stimulated through the grid floor of a footshock box. After 7 days, mice were subjected to a fifth foot shock (1 mA, 4 s), as an integral component of the enhanced-SPS animal model. To increase fear conditioning, mice were re-exposed to the footshock box for 5 min (re-experiencing the past trauma), 1 h after the last administration.

2.3. HP treatment

The HP used in this study was donated by the Korean Pharmacopuncture Institute (Seoul, Korea). The pharmacopuncture treatment method was performed as previously described [14]. Utilizing anatomical reference guides for mice and rats, the ST36 acupoint (Zusanli) was located 5 mm below and lateral to the anterior tubercle of the tibia [15]. The mice received HP treatment bilaterally with doses of $1 \mu\text{L/g}$ or $2 \mu\text{L/g}$. The HP treatment was provided three times per week for 2 weeks. The control and SPSS group received bilateral injections of an equal volume of saline at the ST36 acupoint. The experimental scheme

included HP treatment, behavioral tests, and biological studies (Fig. 1).

2.4. Behavioral tests

2.4.1. Open field test

The Open Field Test (OFT) was conducted as previously described [12]. OFT is widely used to examine the changes in anxiety behavior when mice are exposed to a novel environment [16]. According to the experimental schedule, mice were individually placed in an open field apparatus with black walls and a white floor ($30 \times 30 \times 30 \text{ cm}$), allowing them to freely explore the environment for 20 min. The movements of the mice were recorded using the video-based Ethovision XT System (Noldus Information Technology BV, Wageningen, The Netherlands). The results were analyzed for various parameters, including distance moved, duration of activity, and frequency of specific behaviors within the open field apparatus over a defined time period.

2.4.2. Y-maze test

The Y-maze test was performed following previously established methods [12]. It is widely used to assess spatial learning and memory in rodents. The mice were placed in a Y-shaped maze featuring three identical arms at 120° angles from each other. They were given free access to explore all three arms of the maze. If the animal chose an arm different from the one it initially entered, it was considered an error. Therefore, valid alternating patterns included ABC, BCA, or CAB, while patterns such as CAC or ACA were considered invalid. Over an 8-min duration, both the total number of arm entries and the sequence of entries were recorded. The spontaneous alternation percentage was calculated as follows:

Spontaneous alternation (%) = (number of alternations)/(total arm entries – 2) $\times$ 100.

2.5. Tissue preparation

The experimental procedures involved measuring the body weight of the mice prior to inducing stress and before sacrifice. Anesthesia was induced using avertin (Sigma, St. Louis, MO, USA), following which blood samples were collected from the heart. For western blotting analysis, the hippocampus of the euthanized mice was carefully dissected and stored at -70°C until further use. For immunohistochemistry, euthanized mice underwent transcardiac perfusion with saline to remove blood and preserve the brain. Subsequently, the brains were meticulously removed and post-fixed overnight at 4°C in 4 % paraformaldehyde. The following day, the brains were transferred to a solution containing 30 % sucrose in phosphate-buffered saline and stored at 4°C . Brain sections measuring $30 \mu\text{m}$ in thickness were obtained using a cryostat (Leica Microsystems AG, Jena, Germany) and stored in a storage solution at 4°C until further analysis.

2.6. Corticosterone, glucose, ATP, and glutamate assays

The serum sample preparation followed previously established protocols [12]. Briefly, blood samples were centrifuged at 3000 rpm and then at 12,000 rpm for 10 min each (4°C). The resulting serum samples were stored at -70°C until further use. For the analysis of serum targets, three assays were employed following the manufacturer's instructions: the corticosterone (CORT) assay kit (Biovision, San Francisco, CA, USA), glucose detection kit (Abcam, Cambridge UK), and ATP assay kit (Abcam). Additionally, the glutamate level in the hippocampal region was analyzed using general glutamate assay kits (Abcam) according to the manufacturer's instructions. The optical density (OD) value of each well was measured using a plate reader (Molecular Devices, San Jose, CA, USA) at the absorption wavelength of 450 or 570 nm for each assay.

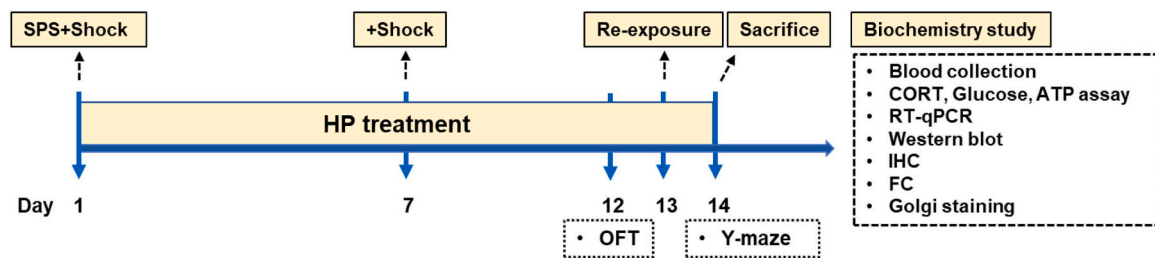


Fig. 1. Experimental scheme. Under the experimental scheme, mice are subjected to stress induction and subsequently undergo behavioral tests. HP treatment is provided three times per week for 2 weeks at the ST36 acupoint. FC, flow cytometry; HP, Hominis placenta; IHC, immunohistochemistry; OFT, open field test; RT-qPCR, real-time quantitative polymerase chain reaction; PTSD, post-traumatic stress disorder; SPSS, single prolonged stress/shock.

2.7. Serum inflammatory cytokines

To determine the concentration of inflammatory cytokines in mouse serum, we analyzed samples using LEGENDplex™ mice Inflammation Panel (13-plex) bead arrays (BioLegend, San Diego, CA, USA) following the manufacturer's instructions. Subsequently, measurements were conducted using LSRFortessa X-20 flow cytometry (BD Biosciences, San Jose, CA, USA).

2.8. Real time-quantitative polymerase chain reaction (RT-qPCR)

The total RNA from the hippocampus was extracted using an RNA extraction kit (Intron Biotechnology, Seongnam-Si, Korea), following a previously described method [12]. Subsequently, total RNA (1 µg) from each sample was synthesized using a cDNA synthesis kit (Bio-Rad Laboratories, Hercules, CA, USA). For qPCR reactions, each sample template was combined with SYBR Supermix (Bio-Rad) and performed using a QuantStudio™ 6 Flex (Life Technologies, Waltham, MA, USA). Data analysis was conducted using the $2^{-\Delta\Delta Ct}$ method based on Ct values normalized to actin. The gene-specific primers used in this study were synthesized by Bioneer Corp. (Daejeon, Korea); their sequences are provided in Table 1.

2.9. Western blotings

Western blotting was conducted following a previously described method [17]. The hippocampus was lysed with RIPA buffer (Biosesang, Gyeonggi-do, Korea) containing phosphatase and protease inhibitors (Thermo Fisher Scientific, Waltham, MA, USA). Sample proteins (20 µg) were separated on Bolt 4–12 % Bis-Tris Plus gels (Thermo Fisher Scientific) and subsequently transferred onto a polyvinylidene fluoride membrane (Bio-Rad). The transferred membranes were then incubated with primary antibodies overnight at 4°C. The following day, the membranes were incubated with the corresponding secondary antibodies (Santa Cruz Biotechnology, CA, USA) for 2 h at room temperature. Blots were visualized using a ChemiDoc Imaging System (Bio-Rad) and quantified using ImageJ software (version 1.46j; National Institutes of Health, Bethesda, MD, USA).

The primary antibodies used were as follows: NMDAR1 (NR1), phospho-NR1 (Ser896) (pNR1 (ser896)), and brain-derived neurotrophic factor (BDNF) (all 1:500; Invitrogen, Cambridge, MA, USA); NMDAR2B (NR2B), phospho-NMDAR2B (Tyr1472), phospho-NMDAR2B (Tyr1336), Tubulin (all 1:1000; Abcam); Cytochrome C (1:1000; Santa Cruz Biotechnology); pmTOR, mTOR, p AKT, AKT, Phospho-JNK (Tyr183/Tyr185) (pJNK), c-Jun N-terminal kinases (JNK) (all 1:1000; Cell Signaling Technology, Danvers, MA, USA).

2.10. Immunohistochemistry

For immunohistochemical staining, a previously described method was used [12]. After blocking, sections were incubated overnight at 4°C with primary antibodies (BDNF [1:500; Invitrogen], c-fos [1:500; Santa

Cruz Biotechnology], glutamate decarboxylase 67 [GAD67] [1:1000; Abcam]). The next day, the sections were incubated in the secondary antibody for 2 hours. Subsequently, for visualization, the sections were treated with an Avidin-Biotin Complex solution (Vector Laboratories, Burlingame, CA, USA) for 1.5 h, followed by incubation in the 3, 3'-diaminobenzidine peroxidase substrate solution (Vector Laboratories) for 5–10 min. To quantify staining, images of the slices were captured using a microscope (Olympus BX53, Tokyo, Japan). For c-fos and GAD67 staining, the number of c-fos positive cells and GAD67 positive cells was counted blindly following a rigorous protocol to ensure an unbiased analysis.

2.11. Golgi staining

Golgi staining was performed as previously described [12], in accordance with the manufacturer's instructions (FD Neuro-Technologies, INC, Columbia, MD, USA). Brains were sectioned using a cryostat in 100-µm coronal sections. For quantifications, we counted the number of dendritic spines in the hippocampal DG region across 10 sections from each mouse, with each section having a 20-µm diameter.

2.12. Statistical analysis

All data are expressed as means ± standard errors of the mean. Statistical analyses were performed using GraphPad Prism software 9.0 (GraphPad, San Diego, CA, USA). The data were analyzed by one-way analysis of variance (ANOVA) with Tukey's multiple comparison test. Statistical significance levels were set at * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.

3. Results

3.1. HP alleviates anxiety and cognitive decline in SPSS-induced PTSD mice

To investigate the effect of HP treatment on PTSD-like behaviors in SPSS-induced PTSD mice, HP treatment was followed three times a week for 2 weeks (Fig. 1). Initially, to verify the establishment of the PTSD animal model, we assessed serum CORT levels and found an increase in the CORT levels in PTSD mice, indicative of stress induction. Significantly elevated CORT levels were observed in the SPSS group compared to the control group. However, HP treatment remarkably reduced the CORT levels compared to those in the SPSS group (Fig. 2A), suggesting that HP treatment restored the CORT levels elevated by PTSD induction.

The typical symptoms of PTSD include anxiety, depression, and cognitive decline. To evaluate the effect of HP treatment on anxiety and cognitive decline in PTSD mice, we conducted the OFT and the Y-maze test. In the OFT, the SPSS group showed a significant decrease in locomotor activity, the indicated by metrics such as total distance moved, immobility, frequency in the center, latency to the first center zone, and time spent in the center. However, HP treatment (1 µL/g) exhibited a greater modulation of these parameters compared to both the SPSS

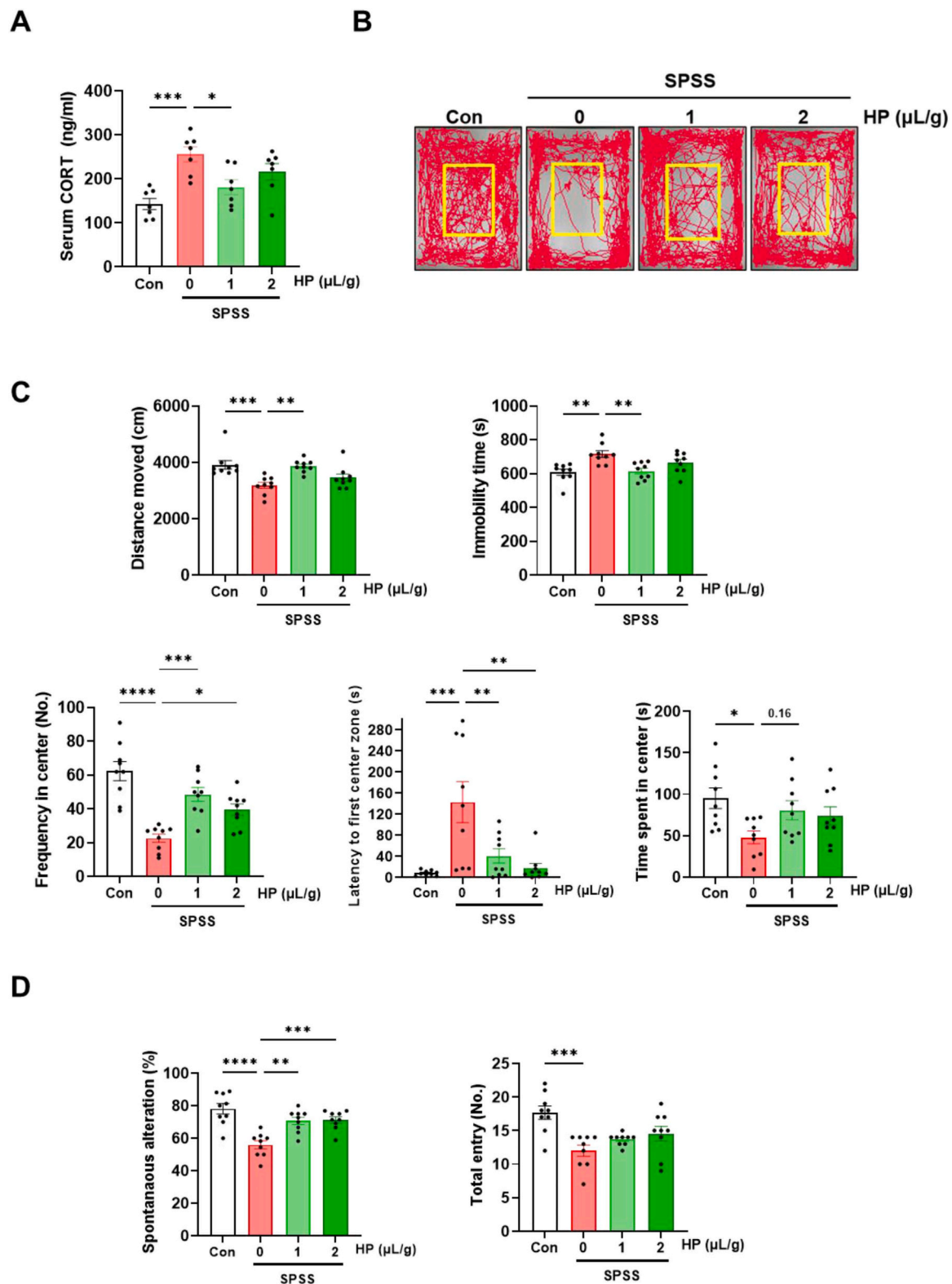


Fig. 2. HP alleviates anxiety and cognitive decline in SPSS-induced PTSD mice. (A) Quantitative analysis of the levels of serum CORT in each group (n=7/group). (B) Representative track plot of distance moved in the open field box. (C) Data represent the results of the OFT, including total distance moved, immobility time, frequency in the center, latency to the first center zone, and time spent in the center (n=9/group). (D) Data depict the results of the Y-maze test, including quantitative spontaneous alteration and total entry (n=9/group). Con, Control mice treated with saline only; CORT, corticosterone; HP, Hominis placenta; SPSS, mice with SPS+Shock treated with saline only; SPSS+HP (1 μL/g), mice with SPS+Shock treated with 1 μL/g HP; SPSS+HP (2 μL/g), mice with SPS+Shock treated with 2 μL/g HP. Each filled circle in the graphs represents an individual mouse. Data are presented as means ± SEMs (**p* < 0.05, ***p* < 0.01, ****p* < 0.001, and *****p* < 0.0001).

group and the 2 $\mu\text{L/g}$ HP group (Fig. 2B–C). Furthermore, cognitive decline was assessed using the Y-maze test, which evaluates hippocampal-dependent short-term and spatial memory in animals. The results showed that spontaneous alternations in the SPSS group were significantly decreased compared to those in the control group. However, HP treatment effectively alleviated spontaneous alternations compared to those in the SPSS group (Fig. 2D). These findings collectively suggest that HP treatment effectively alleviated anxiety and cognitive decline, as shown in hippocampal-dependent behavioral tests, possibly by regulating the CORT levels in PTSD mice.

3.2. HP inhibits inflammation by regulating HPA-axis in SPSS-induced PTSD mice

Given that dysfunction of the HPA-axis is reported as one of the mechanisms underlying the induction of PTSD symptoms, we investigated the impact of HP treatment on HPA-axis dysfunction in PTSD mice [18]. Our results demonstrated that the mRNA expression levels of HPA-axis-related factors, including corticotropin-releasing hormone (CRH), adrenocorticotrophic hormone (ACTH), and glucocorticoid receptor (GR), in the hippocampus were significantly increased in the SPSS group compared to the control group, as determined by RT-qPCR. However, these levels significantly decreased by HP treatment compared to the SPSS group (Fig. 3A). The activation of the HPA-axis is known to be associated with the immune response in PTSD [22]. Additionally, we found that the concentrations of proinflammatory cytokines, such as IL-1 β , TNF- α , IL-6, IFN- γ , and IL-17 in serum, were significantly higher in the SPSS group than in the control group. However, HP treatment significantly decreased these proinflammatory cytokine levels compared to those in the SPSS group (Fig. 3B). Overall, these findings indicate that HP treatment inhibits proinflammatory cytokines by regulating the HPA-axis in SPSS-induced PTSD mice.

3.3. HP treatment attenuates energy metabolism in SPSS-induced PTSD mice

We measured the body weight in each group before stress induction and before sacrifice. The data showed a significant decrease in body weight in the SPSS group compared to the control group, suggesting metabolic dysfunction, whereas HP treatment resulted in increased body weight compared to the SPSS group. Additionally, weight gain was reduced in the SPSS group and partially restored by HP treatment, although not significantly (Fig. 4A). Moreover, the serum levels of glucose and ATP were significantly higher in the SPSS group compared to the control group, indicating glucose metabolism dysfunction, whereas these levels were lower in the HP treatment group (Fig. 4B). Further, we measured the mRNA levels of oxidative phosphorylation

(OXPHOS) related molecules, such as COXI, COXIII, and COXV, in the hippocampus region of PTSD mice using qPCR. The results showed that the mRNA levels of these molecules were significantly increased in the SPSS group compared to those in the control group, indicating mitochondrial dysfunction. However, these levels were decreased by HP treatment compared to those in the SPSS group (Fig. 4C). Furthermore, we investigated the effect of HP treatment on metabolic signaling by conducting western blot analysis to quantify the expression of metabolism-related proteins, such as pmTOR and pAKT. The data showed that they were significantly higher in the SPSS group than in the control group, whereas they were restored by HP treatment compared to the SPSS group (Fig. 4D–E). Collectively, these findings suggest that HP regulates energy metabolism through regulating the AKT/mTOR signaling in PTSD mice.

3.4. HP treatment inhibits glutamate excitotoxicity and regulates glutamate transmission in SPSS-induced PTSD mice

Energy metabolism is essential for maintaining of neuronal signaling and neurotransmission in the brain. First, we measured the amount of excitatory glutamate levels and found that they were increased in the SPSS group compared to those in the control group. However, this increase was restored by HP treatment (Fig. 5A). Further, we assessed the mRNA levels of glutamate transporters, EAAT2 and EAAT3, using qPCR. The results showed that the EAAT2 mRNA levels were significantly lower in the SPSS group compared to those in the control group, whereas EAAT2 mRNA levels were higher in the HP treatment group. Additionally, the levels of EAAT3 were decreased by 27 % in the SPSS group compared to those in the control group, whereas the EAAT3 levels were significantly increased by HP treatment compared to those in the SPSS group (Fig. 5B). Moreover, the levels of phosphorylated NR1 (Ser896)/NR1, phosphorylated NR2B (Tyr1336)/NR2B, and phosphorylated NR2B (Tyr1472)/NR2B were significantly higher in the SPSS group than in the control group, whereas they were significantly lower in the HP treatment group compared to those in the SPSS group (Fig. 5C, D). These results suggest that HP can regulate glutamate reuptake and glutamate transmission. Furthermore, we found that GAD67 positive cells were significantly decreased in the SPSS group compared to the control group, whereas HP treatment recovered GAD67 positive cells (Fig. 5E, F). Furthermore, we measured glutamate excitotoxicity-related markers, namely pJNK and cytochrome C. Their levels were significantly increased in the SPSS group compared to those in the control group, whereas they were lower in the HP treatment group than in the SPSS group (Fig. 5G, H). Moreover, HP treatment inhibits glutamate excitotoxicity by regulating glutamate transmission in PTSD mice.

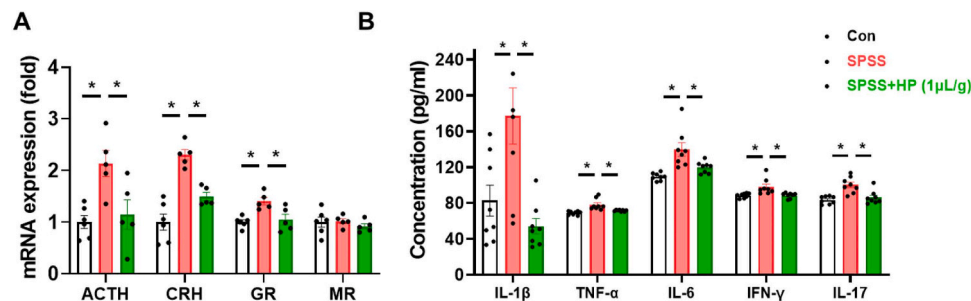


Fig. 3. HP inhibits inflammation by regulating the HPA-axis in SPSS-induced PTSD mice. (A) Quantitative analysis of the relative mRNA expression of ACTH, CRH, GR, and MR in the hippocampus of each group was done using RT-qPCR ($n=5-6/\text{group}$). (B) Quantitative analysis of the levels of interleukin (IL)-1 β , TNF- α , IL-6, IFN- γ , and IL-17 in each group were analyzed using LEGENDplex and flow cytometry ($n=8/\text{group}$). ACTH, adrenocorticotrophic hormone; Con, Control mice treated with saline only; CRH, corticotropin-releasing hormone; GR, glucocorticoid receptor; HP, Hominis placenta; SPSS, mice with SPS+Shock treated with saline only, SPSS+HP (1 $\mu\text{L/g}$); mice with SPS+Shock treated with 1 $\mu\text{L/g}$ HP. Each filled circle in the graphs represents each mouse. Data are presented as means $\pm$ SEMs (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$).

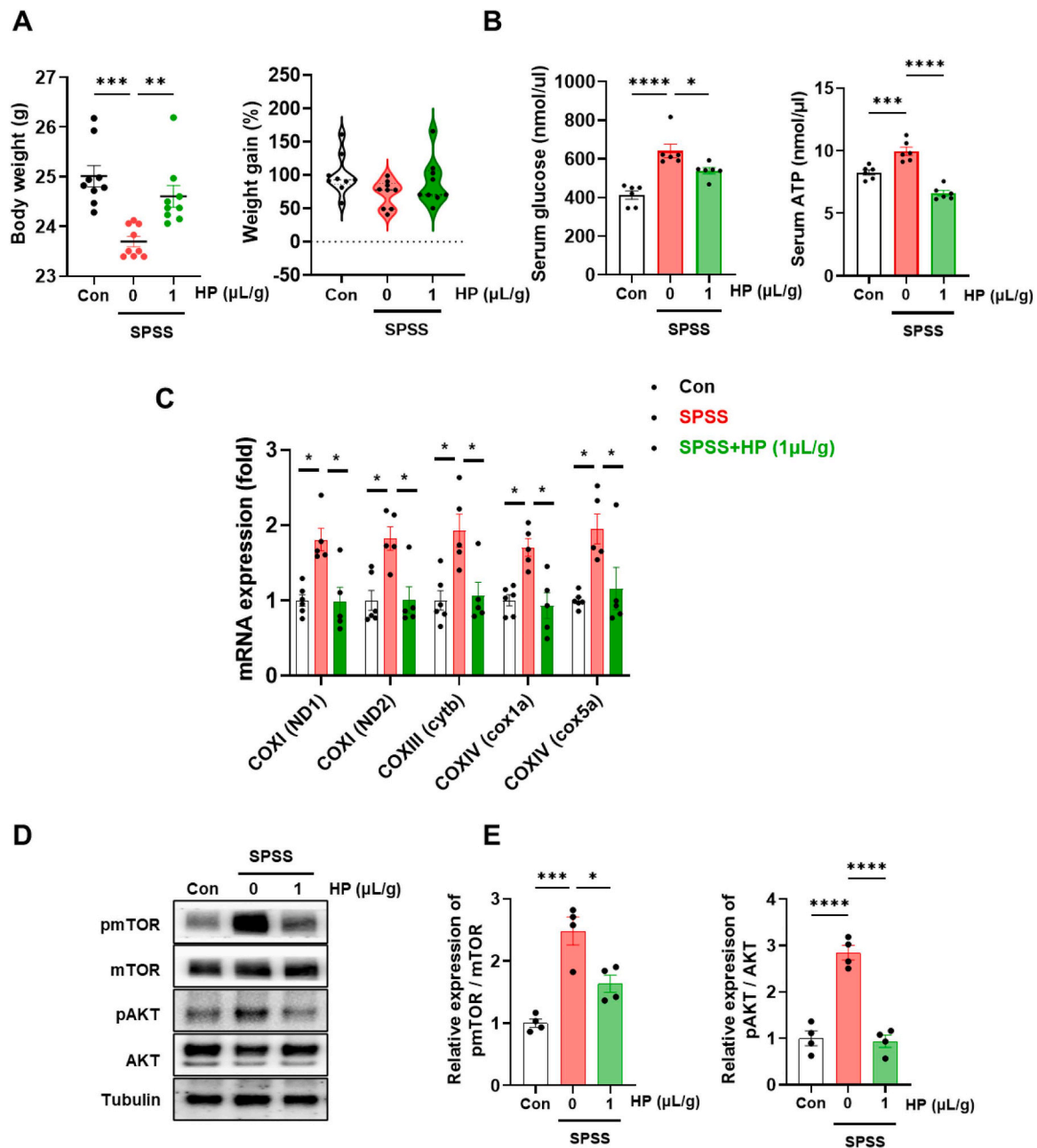


Fig. 4. HP attenuates energy metabolism in SPSS-induced PTSD mice. (A) Data show the results of body weight and weight gain in each group ($n=9/\text{group}$). (B) Quantitative analysis of the levels of serum glucose and ATP in each group ($n=6/\text{group}$). (C) Quantitative analysis of the relative mRNA expression of *COXI(ND1)*, *COXI(ND2)*, *COXIII (cytb)*, *COXIV (cox1a)*, and *COXIV (cox5a)* in the hippocampus of each group were analyzed using RT-qPCR ($n=5-6/\text{group}$). (D) Representative western blot images in each group. (E) Quantitative analysis of the bands showing the levels of pmTOR/mTOR, and pAKT /AKT in each group. Tubulin was used as the loading control ($n = 4/\text{group}$). Con: Control mice treated with saline only, SPSS: mice with SPS+Shock treated with saline only, SPSS+HP (1 $\mu\text{L/g}$): mice with SPS+Shock treated with 1 $\mu\text{L/g}$ HP. Each filled circle in the graphs represents each mouse. Data are presented as the means $\pm$ SEMs (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$).

3.5. HP treatment regulates neuronal activity by modulating BDNF in SPSS-induced PTSD mice

Neuronal activity was assessed using immunohistochemical staining for c-fos positive cells. The number of c-fos positive cells in the hippocampus was significantly decreased in the SPSS group, whereas that of c-fos positive cells increased by HP treatment (Fig. 6A). Next, to visualize dendritic spines, we performed Golgi staining. The results showed a decrease in spine density in SPSS-induced PTSD mice, whereas this was increased in response to the HP treatment (Fig. 6B). The expression of neurotrophic factors is known to alter mood disorders and contribute to

memory decline. Therefore, we investigated the alteration of BDNF, which is involved in emotional and cognitive function. The results showed that the BDNF levels were significantly decreased in the SPSS group, whereas they were restored by HP treatment (Fig. 6C-E). These results demonstrate that HP treatment regulates neuronal activity by modulating BDNF.

4. Discussion

Currently, patients with PTSD are treated with various psychological and pharmacological therapies, including prolonged exposure therapy

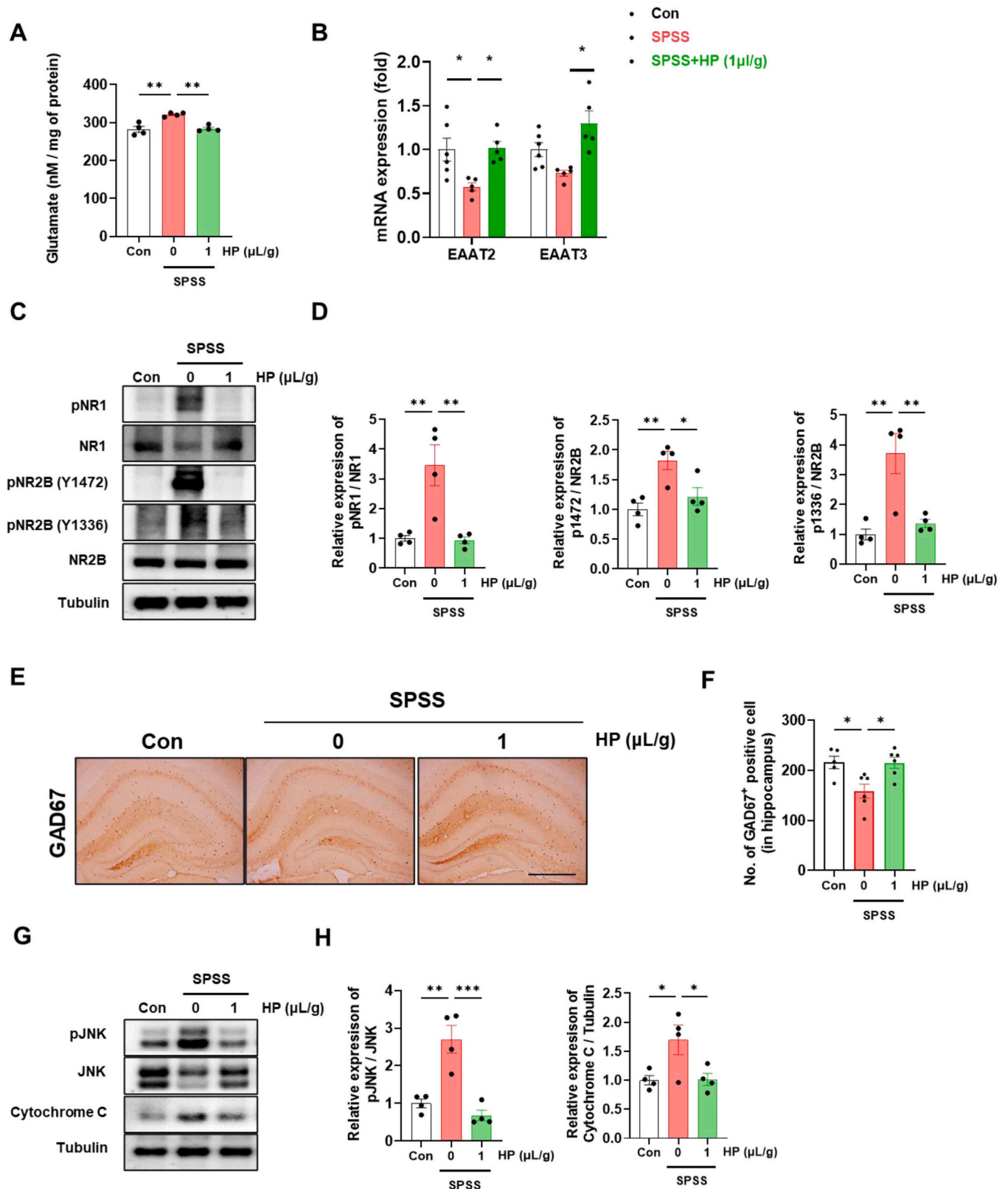


Fig. 5. HP inhibits glutamate excitotoxicity and regulates glutamate transmission in SPSS-induced PTSD mice. (A) Data show the glutamate levels in each group (n=5/group). (B) Quantitative analysis of the relative mRNA expression of *EAAT2* and *EAAT3* in the hippocampus of each group is done using RT-qPCR (n=5-6/group). (C) Representative western blot images in each group. (D) Quantitative analysis of the bands showing the levels of pNR1(ser896)/NR1, pNR2B (Y1472)/NR2B, and pNR2B (Y1336)/NR2B in each group. (E) Representative images of GAD67 immunostaining in the hippocampus for each group. (F) Quantification of the number of GAD67⁺ positive cell in each group (n=5-6/group). Scale bar = 500 μm. (G) Representative western blot images in each group. (H) Quantitative analysis of the bands showing the levels of pJNK/JNK, and Cytochrome C/Tubulin in each group. Tubulin is used as the loading control (n=4/group). Con, Control mice treated with saline only; SPSS, mice with SPS+Shock treated with saline only; SPSS+HP (1 μL/g): mice with SPS+Shock treated with 1 μL/g HP. Each filled circle in the graphs represents each mouse. Data are presented as means ± SEMs (**p* < 0.05, ***p* < 0.01, ****p* < 0.001, and *****p* < 0.0001).

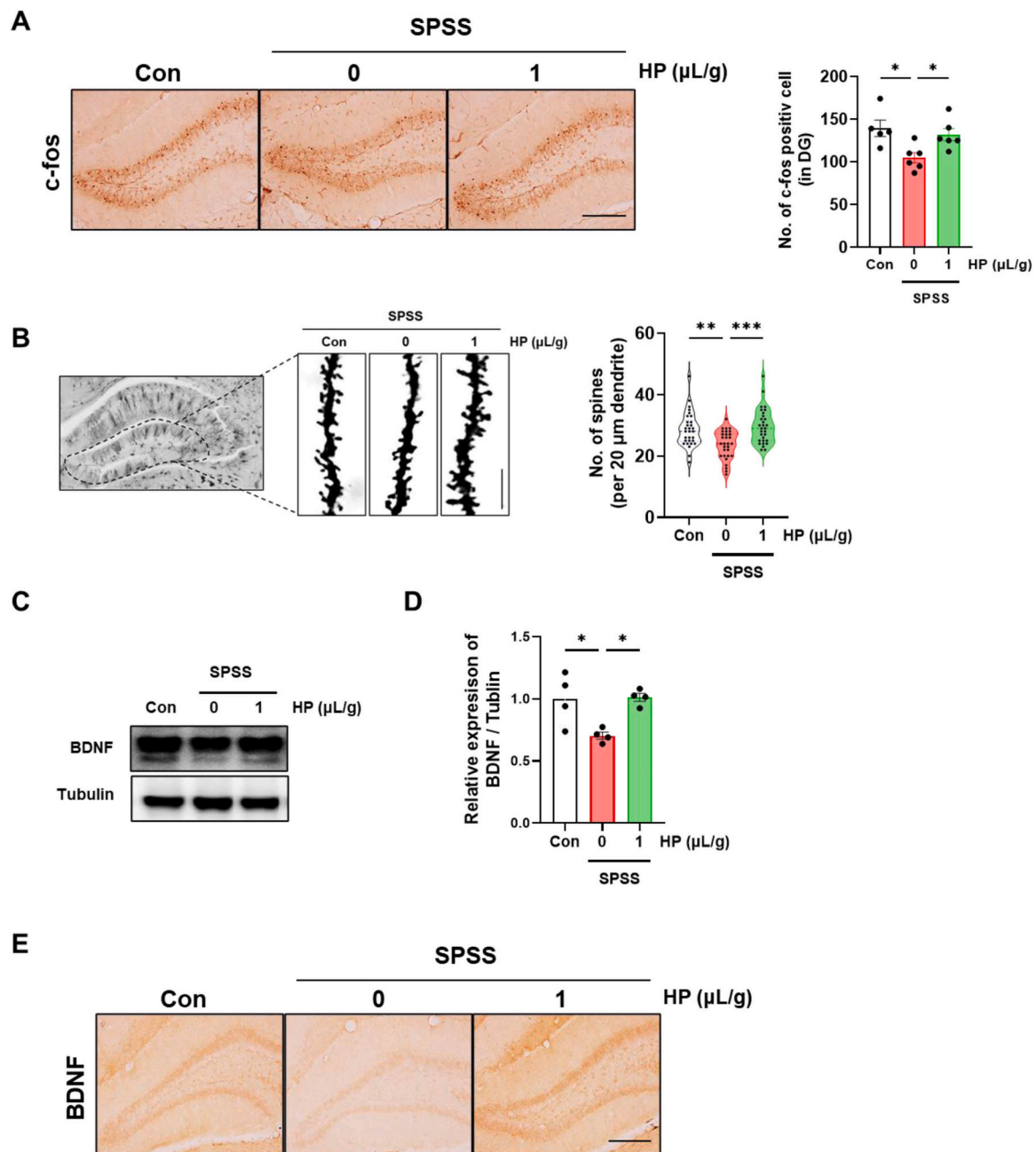


Fig. 6. HP regulates neuronal activity by modulating BDNF in SPSS-induced PTSD mice. (A) Representative images of c-fos immunostaining and quantification of the number of c-fos positive cells in the DG region of the hippocampus for each group ($n=5-6/\text{group}$). Scale bar = 200 μm . (B) Representative images of dendritic spines and quantification of the number of dendritic spines in the DG region of the hippocampus ($n=3/\text{group}$). Each filled circle in the graphs represents an individual dendritic spine of each mouse. Scale bar, 5 μm . (C) Representative western blot images in each group. (D) Quantitative analysis of the bands showing the levels of BDNF/Tubulin in each group. Tubulin is used as the loading control ($n=4/\text{group}$). (E) Representative images of BDNF immunostaining in the DG region of the hippocampus in each group. Scale bar = 200 μm . Con: Control mice treated with saline only, SPSS: mice with SPS+Shock treated with saline only, SPSS+HP (1 $\mu\text{L/g}$): mice with SPS+Shock treated with 1 $\mu\text{L/g}$ HP. Each filled circle in the graphs represents each mouse. Data are presented as means $\pm$ SEMs (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$).

and SSRI drugs. However, there is a demand for more effective and safer treatments to address the needs of patients with PTSD. Complementary and alternative medicine (CAM) therapies, including herbal medicine, acupuncture, and moxibustion, have shown efficacy in such cases. HP is a traditional medicine commonly utilized in East Asia. Moreover, HP treatment has also been shown to be effective in stress-induced alopecia areata in case reports [11]. However, the alleviating effects of HP treatment on anxiety and cognitive decline in PTSD remain unclear. In this study, we examined the effects of HP treatment on PTSD-like behaviors and the underlying molecular mechanisms in the SPSS-induced

PTSD animal model.

As previously described, SPSS is an enhanced SPS model where additional foot shocks are provided following SPS stimulation to better replicate the clinical characteristics of PTSD, including symptomatology (face validity) and pathophysiology (construct validity) [12,19]. PTSD mice exhibited anxiety-like behavior and memory decline, similar to the symptoms observed in individuals with PTSD. We found that locomotor activity was significantly altered in the SPSS group, implying that PTSD mice experienced significant anxiety, as measured by the OFT test. However, HP treatment (1 $\mu\text{L/g}$) attenuated these behavioral elements.

Additionally, the hippocampus plays a crucial role in modulating cognitive function, memory, anxiety, and fear processing [20]. The Y-maze test, which assesses hippocampus-dependent short-term memory test, showed that HP treatment improved spontaneous alternation compared to the SPSS group. These data suggest that HP treatment attenuates PTSD-like behaviors in PTSD mice.

The HPA axis is the primary hormonal mediator of the stress response, especially in PTSD cases [21]. Especially, cortisol, biological marker associated with mental disorder, can play a role in the onset and severity of symptoms related to in the social isolation and anxiety. Once the axis is activated, CRH, ACTH, and CORT are released from the hypothalamus, pituitary gland, and adrenal cortex, respectively. Both GCs and mineralocorticoids activate GR and MR, with GR playing a key role in energy distribution (glycogenesis) and immune function [18]. These finding are consistent with our data showing that HP treatment (1 μ L/g) effectively regulates serum CORT levels and the mRNA levels of *ACTH*, *CRH*, and *GR* in the hippocampus of PTSD mice. Furthermore, PTSD is closely linked to the dysregulation of immune function through the HPA-axis, leading to increased levels of proinflammatory cytokines, such as IL-6 and IL-17. Increased proinflammatory cytokines negatively impact neurogenesis, learning, memory, and synaptic plasticity. Moreover, IL-6 is associated with the occurrence and severity of PTSD symptoms. Our results supported these evidences that HP restored increased proinflammatory cytokine levels in PTSD mice. These findings underscore the crucial role of HP in attenuating PTSD-like behaviors by modulating inflammation and the HPA-axis.

The dysfunction of the HPA-axis can affect various physiological processes, including energy regulation, food intake, and body weight. Additionally, chronic dysfunction of the HPA axis affects glutamatergic signaling in brain regions that regulate emotional behavior [22]. Moreover, dysregulation of glutamate energy metabolism leads to mental disorders and neurodegenerative disease [23,24]. In the current study, body weight was decreased in the SPSS group but restored following HP treatment. Additionally, the levels of glucose and ATP were increased in the SPSS group, indicating hypermetabolism in PTSD, which decreased by HP treatment. In addition, our results demonstrated that HP treatment regulated ATP synthesis-related OXPHOS complexes in the SPSS group. These findings are consistent with previous reports demonstrating that mitochondrial OXPHOS controls brain activity and function via ATP production [25]. Furthermore, MFN2 dysfunction releases mitochondrial cytochrome C into the cytoplasm, inducing mitochondria-dependent apoptosis in neurons [26]. Additionally, disruptions in PI3K-AKT/mTOR signaling is closely linked to the development of various neurodevelopmental disorders. Moreover, mTOR function is regulated by neuronal surface receptors and channels, such as NMDAR, AMPAR, BDNF, which are crucial for the induction and maintenance of LTP and LTD. Our data support this evidence, indicating that HP treatment regulated the AKT/mTOR pathway and reduced Cytochrome C release. Collectively, these findings suggest that targeting the AKT/mTOR pathway to modulate metabolism could offer a novel approach to treating PTSD.

Brain energy metabolism is essential for neurotransmission, neuroplasticity, cell elasticity under stress conditions, and behavior progression. Additionally, glutamate neurotransmission is believed to be pivotal in the pathophysiology of stress-related psychiatric disorders [27]. Excessive accumulation of the extracellular glutamate causes neuronal cell damage and weakness; the regulation of the glutamate uptake in the brain is very important. Our data support the idea that HP treatment modulates glutamate reuptake to regulate the levels of glutamate in the hippocampus of PTSD mice. Therefore, EAAT2 may be a potential target for the treatment of PTSD. NMDAR activation directly correlates with glutamatergic synaptic strength. Calabrese et al. suggested that the activation of pNR1(Ser896) subunit was associated with increased chronic stress [28]. Furthermore, increased synaptic NMDAR activity initiates neuroprotective pathways that promote neuronal survival, whereas elevated extrasynaptic NMDAR activity triggers cell death

pathways, leading to neurodegenerative processes and emotional disorder. Lau et al., demonstrated that BDNF acted as a regulator of the balance between survival (synaptic NMDA receptors) and death (extrasynaptic NMDA receptors), indicated by pNR2B (Tyr1472) and pNR2B (Tyr1336) signaling [29]. In addition, the GABAergic system is involved in the onset of depression, anxiety, and PTSD. Many studies showed that the inhibitory effects of the GABAergic system were reduced in the anxiety model [30]. Taken together, these findings suggest that HP treatment regulate glutamate transmission and GAD67-mediated GABA synthesis to reduce the levels of glutamate. In addition, the dysregulation of HPA-axis and overstimulation of extrasynaptic NR2B-containing NMDAR trigger glutamate excitotoxicity, leading to neuronal damage and weakness, as well as reduced synaptic connectivity owing to the decrease in spine density and dendritic remodeling [27]. These results are consistent with the findings showing that HP inhibits glutamate excitotoxicity-related markers, such as pJNK and Cytochrome C, compared to the SPSS group.

Furthermore, BDNF regulates the balance between survival-promoting synaptic and death-inducing extrasynaptic NMDAR, thereby providing acquired neuroprotection by enhancing synaptic activity and reducing death signaling associated with extrasynaptic NMDAR. As previously described, BDNF modulates neuronal activity, changes in glutamate and GABA levels, affects neuroplasticity in the brain, and regulates emotional and cognitive function [31]. Our data support this evidence, showing that the number of c-fos positive cells was regulated by the HP treatment. Additionally, the morphology and density changes of dendritic spines reflect alterations in neuronal plasticity (LTP and LTD) and neural activity. Bennett et al. suggested that BDNF controlled dendritic spine formation and regression in response to the stress and trauma [32]. These observations suggest that BDNF modulates neuronal activity, resulting in increased glutamate levels and the induction of glutamate excitotoxicity in PTSD mice. Therefore, HP treatment improves neuronal activity and glutamate transmission in PTSD mice, targeting BDNF may serve as a potential therapeutic approach for the treatment of PTSD.

In future investigations, we aim to identify the bioactive compounds present in the HP, thereby elucidating more specific molecular mechanisms underlying the efficacy of HP in treating PTSD. Furthermore, the impact of HP treatment on fear extinction warrants exploration in future studies. Additionally, clinical trials are necessary to evaluate the efficacy of HP for PTSD.

5. Conclusions

In summary, HP treatment effectively attenuated PTSD-like behaviors, including anxiety and cognitive impairments, by modulating energy metabolism and neuronal activity in PTSD mice. These findings suggest that HP treatment could be a promising therapeutic intervention for PTSD.

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CRediT authorship contribution statement

Mudan Cai: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Hee Ra Park:** Investigation. **Eun jin Yang:** Writing – review & editing, Project administration, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

The authors did not utilize any generative AI or AI-assisted

technologies during the preparation of this work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.biopha.2024.117243](https://doi.org/10.1016/j.biopha.2024.117243).

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